

Supply Chain Stockout Prediction

End-to-End Deep Learning Approaches

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1. The Stockout Prediction Problem

- **Business Objective:** Accurately predict future out-of-stock events to optimize supply chain operations and prevent lost sales.
- **The Challenge:** Moving from simple static backorder prediction to dynamic, time-series stockout forecasting.
- **Core Task:** Predict whether a specific product (SKU) at a specific store will experience a stockout in the future (next 24 hours vs. next 7 days).

2. Dataset & Signal Extraction

- **Dataset Selection:** FreshRetailNet-50K
 - Provides true time-series daily records at the SKU-Store level over 97 days.
 - Contains hourly stock-status tracking (predicting stockouts during active hours: 6 AM to 10 PM).
- **Targeting High-Impact SKUs:**
 - Evaluated the Top 15 SKUs statistically based on high sales volume and high stockout variance.
 - Focuses the models on the most challenging, non-trivial demand patterns.

3. Establishing a Baseline

- **Model:** PyTorch Tabular Transformer
- **Approach:** Treats daily features (like sales, discounts, weather) as a sequence of categorical and numeric embeddings.
- **Baseline Results:**
 - Accuracy: 55.83% | AUC: 0.5257
- **Key Takeaway:**
 - Barely outperformed random guessing (0.50).
 - Proved that treating stockout prediction as a simple classification problem is insufficient. Sequential time-series modeling and feature engineering are strictly required.

4. Short-Term Forecasting (Next 24 Hours)

- **Objective:** Predict next-day hourly stock status using recent history (7 to 14 days).
- **RNN Experimentation:**
 - Tested various RNN architectures; a SimpleRNN (32 units) achieved **75.88% Accuracy**.
 - Complex architectures degraded due to noise and underfitting.
- **LSTM Ablation:**
 - Standard LSTM (44k params) achieved 0.785 F1-score.
 - A **Compact Feature-Reduced LSTM** (6 features, 5.9k params) outperformed it with a **0.790 F1-score**, reducing parameters by 87%.

4.1. Exploring the Ultra-Compact LSTM Limit

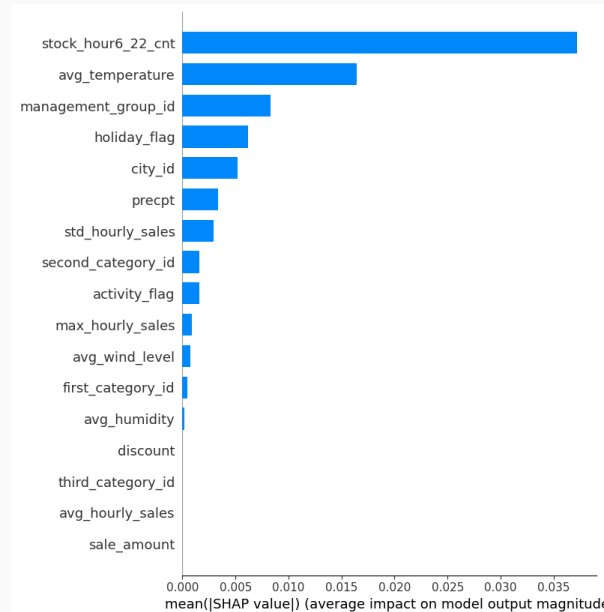
- **Goal:** Test the lower bound of LSTM complexity to see if even smaller models are viable.
- **Ultra-Compact Variants:**
 - **h16 (6 features):** 1,944 params | F1: 0.741 (Drops F1 significantly)
 - **h8 (6 features):** 728 params | F1: 0.647 | Recall: 0.924 (Recall-heavy, poor balanced F1)
 - **h16 (4 features):** 1,816 params | F1: 0.746
- **Findings:**
 - Shrinking beyond h32 trades off too much F1-score for parameter reduction.
 - The h32 model (5,912 params) remains the optimal **practical minimum**.
 - **Recommendation:** Use h32 as the primary LSTM, retaining h16 only as an ultra-light fallback.

5. Long-Term Forecasting (7-Day Horizon)

- **Objective:** Predict 7-day future stockouts using a 28-day historical window.
- **The Breakthrough:**
 - Engineered time-aware features: `stockout_streak` and `stockout_rolling_7`.
- **Model Rankings:**
 - 🥇 **LSTM Seq2Seq:** Achieved Best pure ranking (PR-AUC 0.5613), crossing the naive persistence baseline.
 - 🥈 **GRU Attention:** Achieved Highest F1-Score (0.5201).
- **Conclusion:** Sequence-to-Sequence models with engineered temporal features are highly effective for extended stockout forecasting horizons.

6. Explaining the Stockout Triggers (SHAP)

- Supply chain managers need to trust the model. We used **SHAP** to pinpoint exactly which features (e.g., temperature spikes or recent promotions) triggered a short-term stockout alert.



7. Explaining Temporal Importance (Attention)

- **Temporal Attention:** Visualizes **when** the LSTM Seq2Seq model looks back into the 28-day history (X-axis) to predict a future stockout day (Y-axis).

